EEC-204: Tutorial - I
Control Systems

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1. Using residue method for partial fraction expansion, find the signals whose Laplace transforms are given by:

   a) $X(s) = \frac{2s}{(s+2)(s-2)}$
   b) $X(s) = \frac{2s+1}{s^2(s+1)^2}$
   c) $X(s) = \frac{s^3+2s^2+2s+5}{s^2(s^2+2s+5)}$
   d) $X(s) = \frac{4s-8}{(s^2+4)^2}$.

2. a) Consider a signal with Laplace transform:

   $$X(s) = \frac{s^2+4}{s^3(s^2+2s+5)^2}$$

   Give the form of $x(t)$ that would result from a partial fraction expansion. Express the signal as a linear combination of time functions, but do not solve for the coefficients themselves.

   b) Repeat (a) but for:

   $$X(s) = \frac{s-1}{(s+2)^4(s-3)^3(s+4)}$$

3. For the signals whose Laplace transforms are given below, indicate whether the signals are bounded and, if so, whether $\lim_{t\to\infty} x(t)$ exists. If the limit exists, give its value. Do not invert the Laplace transforms to obtain the results.

   a) $X(s) = \frac{10}{(s^2+1)^{10}}$.
   b) $X(s) = \frac{(s-1)}{s(s+2)}$.
   c) $X(s) = \frac{1}{s^2(s+2)}$.
   d) $X(s) = \frac{5}{s(s+1)^2}$
   e) $X(s) = \frac{2(s-1)}{(s^2+2s+1)(s+3)}$
   f) $X(s) = \frac{2(s-1)}{(s^2+2s+2)(s+3)}$
   g) $X(s) = \frac{2(s-1)}{(s^2+2s+2)^2(s+3)}$

4. a) Give the response $y(t)$ of a system with transfer function:

   $$P(s) = \frac{1}{s^2(s+1)}$$

   and with constant input $x(t) = 1$.

b) An input $x(t) = sin(2t)$ is applied to a system with transfer function:

$$P(s) = \frac{s-1}{s(s^2+4s+8)^3(s^2+4)}$$

List all the functions that appear in the partial fraction expansion of $Y(s)$, without calculating the coefficients of the expansion. Express the result in terms of real functions, and indicate which functions must have nonzero coefficients.

c) Give the real time function $x(t)$ whose transform is:

$$X(s) = \frac{1+2j}{(s+2+3j)^3} + \frac{1-2j}{(s+2-3j)^3}$$